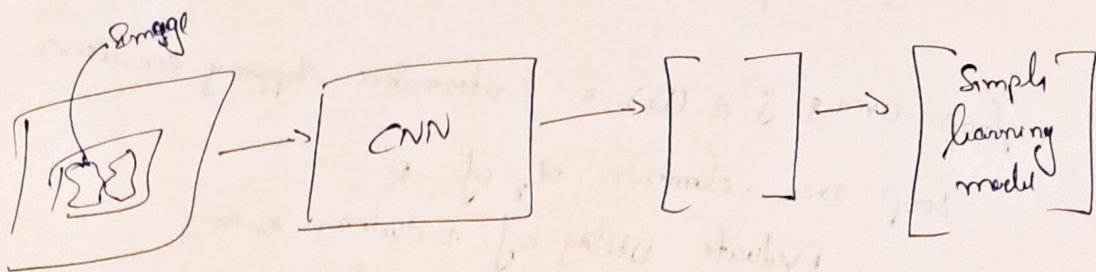


Feature from pretrained model



Feature Reduction

$$X \in \mathbb{R}^{D \times 1} \rightarrow \hat{x} \in \mathbb{R}^{d \times 1} \quad d < D$$

Problem: Best subset of size d out of D

Methods: i) Filter based: Examine each feature individually

Classification:

Loop over: [Evaluate utility of just using x_i]

Select top- d subset

Retain on top d

]

Limitation: a single x_i may not give good classification, but in combination with other may give good classification

ii) wrapper based method

- (i) Forward selection
- (ii) Backward elimination

Forward selection

$$S \leftarrow \emptyset ; R \leftarrow \{x_1, \dots, x_D\}$$

loop over $j \in (D)$ + (alternative stopping conditions)

loop over elements x_j of R

Evaluate utility of including x_j in S

Pick the best feature, included in S

Remove from R

time $O(D^2)$

ii) Backward elimination

$$S \leftarrow \{x_1, x_2, \dots, x_D\} ; R \leftarrow \emptyset$$

loop over $j \in (D)$ + alternative stopping conditions

loop over elements x_j of S

Evaluate utility of removing x_j from S

Remove the worst feature from S add to R

3) Embedded method:

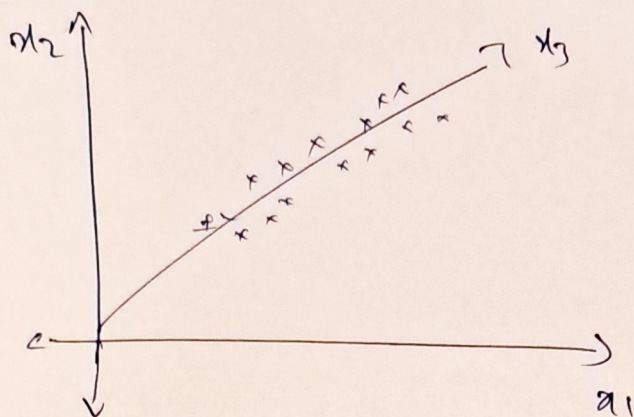
example: Lasso [L1 regularization]
Elastic Net ($L_1 + L_2$)

* Learning and Subset Selection are integrated

$$L_2: \text{Sum} = \frac{\lambda_2}{2} \|w\|_2^2 + \sum [C(1 - t|y_i|)]$$

$$L_1: \text{Sum} = \lambda_1 \|w\|_1 + \sum [C(1 - t|y_i|)]$$

PCA (Principal Component Analysis)



$$x_3 = ax_1 + bx_2$$

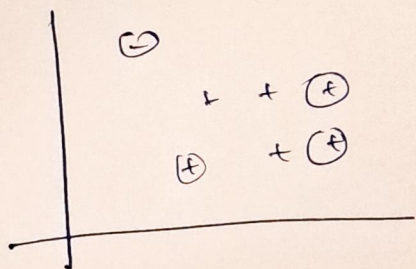
- PCA is feature reduction
- Previous method are feature elimination

$y = f(w_1x_1 + w_2x_2 + \dots + w_nx_n)$
 normalization → If gaussian assumed
 Std mean = 0, $\sigma = 1$

If assumed uniform non-mon normalised

Kernelised SVM

Kernel: way to compare 2 samples in similarity



$K(x_1, x_2) \rightarrow \text{high}$
 $K(x_1, \tilde{x}) \rightarrow \text{low}$